

Electroencephalogram and Event-Related Potential in Mild Cognitive Impairment: Recent Developments in Signal Processing, Machine Learning, and Deep Learning

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Abstract—Mild cognitive impairment (MCI) is an early stage of non-age-related cognitive decline with an increased risk of progressing to dementia. Early detection of MCI is essential for implementing preventative strategies that can delay or prevent the onset of dementia, ultimately improving patient outcomes and reducing healthcare costs. Electroencephalograms (EEGs) and event-related potentials (ERPs) have shown significant promise in detecting MCI due to their affordability, real-time monitoring capabilities, and noninvasiveness. EEG provides continuous brain activity data, while ERPs offer insights into specific cognitive processes by analyzing brain responses to stimuli.

These methods can complement each other in MCI diagnosis by providing a comprehensive view of overall brain function and detailed information on specific cognitive processes. However, EEG and ERP are susceptible to noise and interindividual variability, which can hinder their reliability. In addition, applying machine learning models on EEG or ERP for MCI detection presents challenges such as the risk of overfitting and difficulties in interpreting the underlying decision-making process. This review emphasizes recent advancements in signal processing and feature extraction methods applied to EEG and ERP data and explores the use of machine learning and deep learning techniques to enhance diagnostic accuracy and interpretative depth. By integrating these methodologies, the review highlights how EEG and ERP can contribute to a more effective understanding and monitoring of cognitive changes associated with MCI, underscoring the importance of early diagnosis for timely intervention and improved patient care. Finally, the review focuses on future research directions, including the development of advanced analytical techniques and multimodal integration approaches involving EEG and ERP to further improve diagnostic accuracy and clinical application.

Index Terms—Deep learning, electroencephalogram (EEG), event-related potentials (ERPs), machine learning, mild cognitive impairment (MCI), signal processing.

I. INTRODUCTION

MILD cognitive impairment (MCI) is broadly defined as an intermediate stage of cognitive decline between normal aging and dementia. MCI patients are at higher risk of developing dementia [1], [2]. Nevertheless, not all individuals with MCI will transition to Alzheimer's disease (AD) and dementia [3], [4]. Identifying biological markers that differentiate MCI from healthy aging is essential for developing preventative interventions, which can enhance quality of life, reduce caregiver burden, and lower the cost of dementia care [5], [6].

To better understand and monitor these cognitive changes, electroencephalogram (EEG) and event-related potential (ERP) have shown significant potential [7], [8]. Compared to the diagnostic biomarkers of AD, such as magnetic resonance imaging

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(MRI) scans, cerebrospinal fluid (CSF), and positron emission tomography (PET) scans, EEG offers several advantages [8], [9]:

- 1) *Cost-effectiveness*: EEG is relatively inexpensive and more accessible than MRI, CSF analysis, or PET scans, making it a practical option for widespread screening.
- 2) *Real-time monitoring and portability*: Unlike MRI and PET scans, which provide static images, EEG, specifically using portable EEG devices, with high temporal resolution offers real-time data on brain activity, allowing for dynamic observation of neural processes and changes over time.
- 3) *Noninvasiveness*: EEG, unlike CSF, is a noninvasive technique and does not imply any injections or complex procedures analysis.
- 4) *Complementary insights*: EEG can complement MRI, CSF, and PET scans by providing additional information on brain function and connectivity, enhancing the overall diagnostic accuracy for MCI and AD.

This review aims to provide a comprehensive synthesis of recent advancements in the use of EEG and ERP technologies for the detection and classification of MCI. By critically examining signal processing techniques, feature extraction methods, and applications of machine learning and deep learning, this article aims to highlight the strengths, limitations, and emerging trends in the field. Furthermore, the review seeks to identify gaps in current research and propose future directions, including the integration of EEG and ERP with other neuroimaging modalities and the development of explainable AI approaches to improve the diagnostic accuracy and clinical utility of these noninvasive tools.

In this article, Section II explores the existing methods and challenges in diagnosing MCI, offering insights into contemporary clinical practices. Section III includes the method part of the article. Section IV examines the comparative effectiveness of resting-state and task-state EEG approaches in identifying MCI, highlighting their respective advantages and limitations. Section V delves into the advanced signal processing techniques and neural markers derived from EEG data that are instrumental in detecting and understanding MCI. Section VI focuses on the application of machine learning algorithms to classify MCI and validate the effectiveness of these models in distinguishing between different cognitive states. Section VII investigates the role of deep learning (techniques in enhancing the accuracy and depth of MCI detection, presenting novel approaches and future directions for research. Finally, Section VIII focuses on future research directions, including multivariate multiscale entropy analysis, component differentiation, higher-order interactions (HOIs), causal and explainable deep learning, and multimodal integration of EEG with MRI, functional MRI (fMRI), or functional near-infrared spectroscopy (fNIRS) for MCI.

II. CURRENT CLINICAL PRACTICE FOR MILD COGNITIVE IMPAIRMENT

Deficits in executive function [10], [11] and working memory [12], [13] contribute to the development of MCI and AD. Executive function involves cognitive processes responsible for managing and regulating higher-level mental tasks such as problem-solving and decision-making. Working memory is the cognitive system accountable for temporarily holding and

updating information essential for tasks such as reasoning and learning. Clinical practice for diagnosing MCI uses the clinical criteria for mild neurocognitive disorder in the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition [14] (see Supplementary Material for details about the diagnostic criteria) [15]. Patients can be diagnosed with single or multidomain MCI. Single-domain MCI refers to deficits in only one of the cognitive domains, and multidomain MCI refers to deficits in more than one of these domains. Cognitive domains include processing speed, working memory, executive functioning, verbal memory, visual memory, and language (see Supplementary Material for details about the cognitive test). Experts also classify MCI into two subtypes based on the presence or absence of memory impairments: amnesic MCI (a-MCI) and nonamnesic MCI (na-MCI). Patients diagnosed with a-MCI have a higher probability of progression to AD compared to na-MCI [16]. Particularly, studies showed that the progression rate to severe cognitive impairment is approximately 19.8% for amnesic multidomain MCI, compared to 1.1% for nonamnesic single-domain MCI [17].

In addition to clinical and cognitive assessment, biomarkers are increasingly used to diagnose MCI and dementia [18], [19], [20]. Biomarkers serve as objective indicators to detect biological changes associated with MCI and the underlying neurodegenerative mechanisms. For example, pathological biomarkers of AD include the accumulation of amyloid-beta ($A\beta$) plaques and the formation of neurofibrillary tangles (NFTs) [21]. $A\beta$ plaques consist of abnormal clusters of beta-amyloid protein fragments that build up between nerve cells, while NFTs are twisted fibers made of abnormal tau protein found within nerve cells [21]. These markers have been found in postmortem brain tissue of patients with AD [22] and have been used to diagnose MCI and predict AD. For example, it has been shown that plasma β -amyloid $A\beta_{42/40}$ ratio (TP42/40) is significantly lower in a-MCI patients compared to cognitively normal participants, and this lower ratio increases the risk of AD progression by 70% [23]. These markers can be measured using blood samples, imaging methods (e.g., PET), and CSF [24], [25], [26], [27]. MCI has also been shown to be associated with brain structural changes such as reduced hippocampal volume [20], [28], cortical thickness [29], [30], and abnormal white matter tract fractional anisotropy [31] that can be measured using MRI. For example, by comparing different subtypes of MCI with cognitively normal participants, it has been shown that the atrophy presents focally at entorhinal and parahippocampal cortices in the subtle MCI subtype, and it progresses to cortical thinning in temporal, parietal, and frontal regions in a-MCI patients [29].

Furthermore, genetics can serve as a valuable biomarker for MCI. Studies have found that the ApoE $\epsilon 4$ allele is a risk factor for MCI progressing to AD. Individuals with MCI who carry the ApoE-4 may have a higher likelihood of experiencing cognitive decline and transitioning to AD compared to those without it [32], [33].

At the neurophysiological level, EEG is a noninvasive and inexpensive tool for revealing neurophysiological changes associated with MCI. Brain activity can be recorded while performing cognitive tasks and abnormal activities can represent the mech-

anism of cognitive impairment. For example, EEG is utilized to record prefrontal cortex (PFC) activity as it is the critical region in executive functioning and working memory [34], [35], [36], [37], [38]. PFC shows hyperactivity in MCI patients, which indicates its compensatory mechanism [39]. Hyperactivity in PFC can be measured using functional neuroimaging and EEG. This hyperactivity could indicate compensatory mechanisms with evidence suggesting that this enhanced activation is in response to impairment and is linked to a beneficial effect on cognitive performance [40], [39], [41]. Since changes in EEG patterns often occur before clinical symptoms become apparent, it could help in the early detection of cognitive decline and dementia [41]. Moreover, promoting compensatory mechanisms in MCI patients can delay or prevent developing AD [10], [42], [43]. Cognitive compensation is when the brain adjusts and utilizes different strategies to “compensate” cognitive deficits or obstacles, typically aiming to maintain or enhance cognitive abilities [40].

Cognitive reserve (CR) is another notion suggesting that regularly engaging in intellectually stimulating activities over one’s lifetime alters the brain in a manner that mitigates the adverse impact of brain pathology on cognitive function. CR operates like a “reserve” of mental capabilities, ready to step in and compensate for any reduction or limitation in cognitive functioning [44]. Over a 12-year follow-up study, it has been shown that CR is associated with delayed progression to MCI and cognitive decline [45]. EEG is instrumental in assessing CR and compensation as it reveals changes in brain activity that may precede cognitive and behavioral impairments [46]. Section V discusses EEG features that have been linked to CR and compensatory mechanisms.

III. METHOD

To gather various contributions in the field of EEG- and ERP-based MCI detection, we searched the SCOPUS, Elsevier, and Google Scholar databases using the following keywords.

- 1) EEG or ERP.
- 2) MCI.
- 3) Detection.
- 4) Classification.

These search terms were combined using the following rule: 1) AND 2) AND [3) OR 4)], and the search focused on the title, abstract, or keywords of the papers. We limited our search to English peer-reviewed journal articles published between January 2011 and May 2024. The collected papers were then carefully reviewed to assess their eligibility. To focus exclusively on distinguishing between MCI and healthy controls (HC) based on EEG/ERP, we excluded the following types of papers:

- 1) those focused only on classification between MCI and other illnesses (e.g., subjective cognitive decline, vascular dementia, and AD);
- 2) those performed only multiway classification between MCI, other illnesses, and HC;
- 3) those reporting classification results based on a combination of EEG and other modalities (e.g., MRI, PET, and heart rate variability);

- 4) those with a sample size of fewer than 20 participants. All authors discussed and agreed on the final selection of papers.

IV. RESTING-STATE EEGS VERSUS TASK-STATE EEGS IN MCI DETECTION

An important issue about EEG-based MCI detection is the mental states from which EEG features are extracted. Although a large body of literature focuses on the resting states, some studies have started to pinpoint the potential of task-state EEGs. In this section, we will discuss the critical difference behind the resting-state versus task-state EEGs in terms of MCI detection and conclude with a summary figure to elucidate the current position of both directions, which hopefully will help inform and inspire future research.

Resting-state EEGs refer to the EEGs recorded while participants are at rest, without performing any specific task [47], [48], [49], [50], [51], [52]. Typically, resting-state EEGs can be recorded either with an eye-opened or eye-closed condition. However, there is a higher tendency to induce drowsiness in participants during recording in an eye-closed condition [53]. Therefore, an eye-opened condition is more commonly adopted. Resting-state EEGs allow for the measurement of spontaneous brain activity, which is thought to reflect the intrinsic functional organization and connectivity of the brain [52], [54], [55], [56], [57]. In contrast, task-state EEGs refer to EEGs collected while a participant is engaged in a particular task, which can provide valuable insights into the brain’s functional dynamics during task performance [50], [58], [59], [60].

To further elaborate on the difference between resting-state and task-induced EEGs in classifying MCI, consider this analogy: Imagine we are identifying features in cardiopulmonary signals to classify individuals with weak versus normal cardiovascular endurance. One approach is to analyze features from these signals while participants are at rest. There could be latent “trait-like” features that inherently indicate cardiovascular endurance strength. Alternatively, we can examine features from cardiopulmonary signals during or after a 5K run. This stress test, although more effortful to record, may provide a more targeted indication of cardiovascular endurance strength.

A. Advantages and Limitations of Resting-State EEGs

One of the key advantages of using resting-state EEGs for classifying between individuals MCI and HC is the relatively ease data-collection procedures. Resting-state recordings do not require participants to perform any specific task, which can be particularly beneficial for older adults or those with cognitive impairments who may have difficulty engaging with complex experimental paradigms [61], [62], [63], [64], [65]. In addition, resting-state EEGs can assess global brain condition independent of any cognitive processes engaged during a particular task [48], [50], [66], [67], [68], [69], [70]. Furthermore, resting-state EEG data can be obtained in a shorter time frame than task-based protocols, making it a more time-efficient approach for clinical applications [71].

Despite the advantages of resting-state EEGs, there are also potential limitations that should be considered. Resting-state

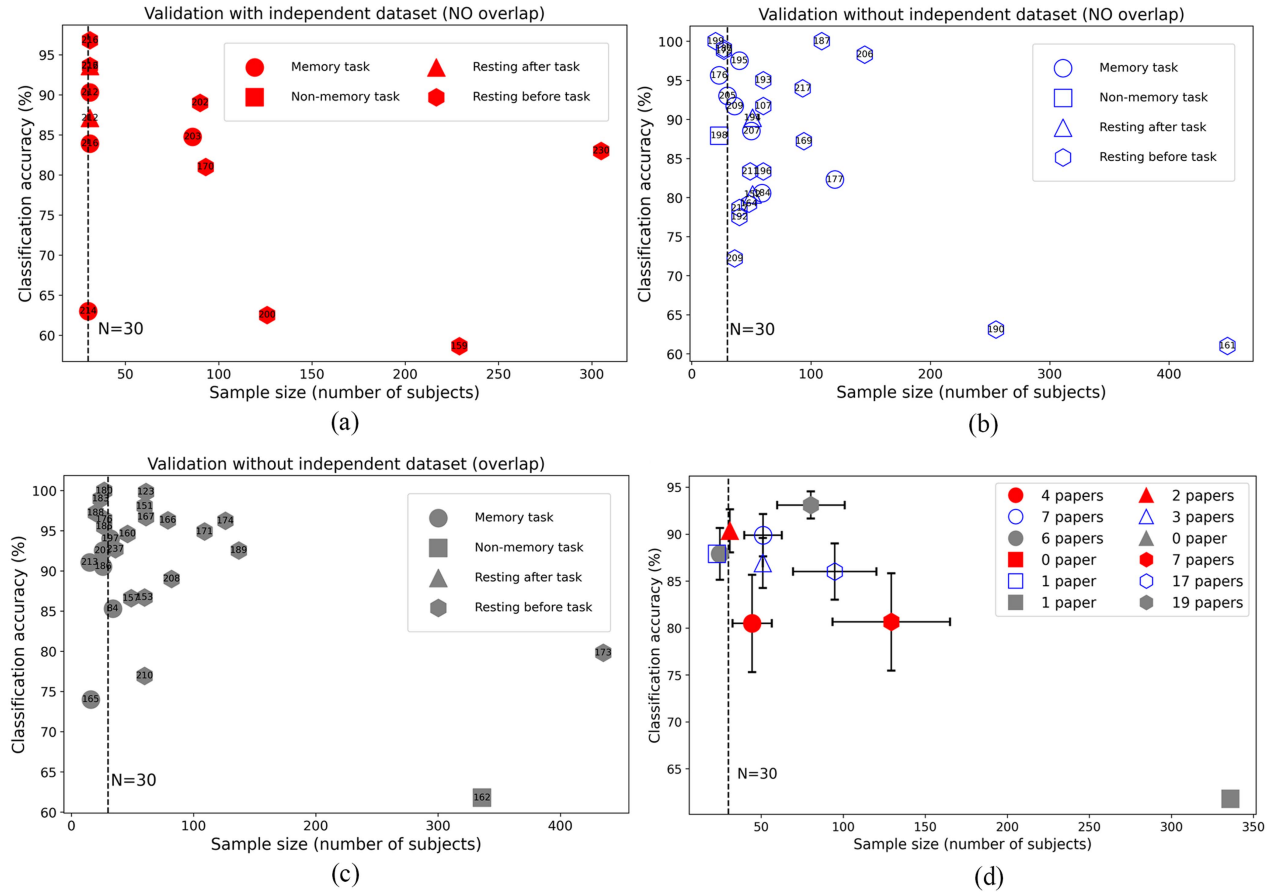


Fig. 1. Graphic summary of classification performances of MCI detection using resting-state versus task-state EEGs. Here we plot the distribution of classification accuracies against study sample size in a nonexhausted list of current literature that uses validation approach (a) with independent dataset (no overlap), (b) without independent dataset (no overlap), and (c) without independent dataset (overlap). Studies were further categorized by the types of EEGs being used, including: memory-task EEGs (circles), nonmemory task EEGs (square), resting-state EEGs after task (triangle), and resting-state before task (hexagon). (d) We further plot the mean and standard deviation of classification performances and study sample size for all combinations of EEG types and validation approaches. The numbers indicate the reference number.

brain activity can be influenced by various factors, such as vigilance, arousal, and intrinsic mental processes, which may introduce more variability into the data compared to task-based EEGs [47], [71], [72], [73]. In addition, interpreting resting-state EEG features in the context of cognitive impairment can be more challenging, as the specific neural mechanisms underlying these measures may not be as well understood as those associated with task-evoked brain responses [47], [71], [72], [73]. Furthermore, the clinical utility of resting-state EEGs for MCI classification may be limited because the observed changes in brain activity may not be specific to MCI but rather reflect more general cognitive decline or neurological alterations [48], [50], [74].

B. Advantages and Limitations of Task-State EEGs

One key advantage of task-state EEGs for MCI classification is their ability to provide a more focused examination of specific cognitive functions that may be impaired in individuals with MCI. Task-evoked brain responses, such as ERPs or oscillatory patterns, can be directly linked to well-characterized neural mechanisms underlying cognitive processes such as memory, attention, or executive function [75], [76], [77], [78], [79].

These task-specific brain signatures may offer more sensitive and specific markers of cognitive impairment than resting-state measures, reflecting a more global assessment of brain activity. Several studies have demonstrated the utility of task-state EEGs for accurately classifying individuals with MCI from HC, with promising classification accuracies reported in the literature. Examples of tasks that have been used include memory tasks [80], [81], [82], attention tasks [78], [83], [84], [85], and executive function tasks [79], among others. Furthermore, task-state EEGs can help elucidate the neurophysiological underpinnings of cognitive deficits in MCI, as the observed changes in task-evoked brain responses can be interpreted in the context of the specific cognitive demands and neural processes engaged during the task [79], [86]. This can provide valuable insights into the neural mechanisms driving cognitive decline and potentially inform the development of targeted interventions or rehabilitation strategies [5], [87], [88], [89], [90].

While task-state EEGs offer valuable insights into the neural correlates of cognitive functions, they also have some potential limitations. The need for subjects to engage in specific cognitive tasks can be challenging for individuals with MCI, who may struggle with task performance or have difficulty following

complex instructions. In addition, the task-specific nature of the brain responses observed in task-state EEGs may limit the generalizability of the findings as the observed changes may not necessarily reflect broader alterations in brain function that are characteristic of MCI. Moreover, the administration of task-based EEG protocols can be more time-consuming and resource-intensive than resting-state recordings, which may hinder their feasibility for large-scale clinical applications. Finally, task-based EEGs are more affected and susceptible to noise from muscles and eye movements.

C. Classification Performances of Resting-State EEGs Versus Task-State EEGs

Fig. 1 provides a graphic summary of the classification accuracy based on EEGs from different states from a nonexhausted list of publications (for articles inclusion criteria, refer to Section I). First, we focus on EEGs from four states:

- 1) *resting-state EEGs before task*: recorded during task-free baseline;
 - 2) *resting-state EEGs after task*: recorded after participants perform certain tasks;
 - 3) *memory task EEGs*: recorded during participants performing a memory-relevant task (e.g., a delay-match-to-sample task);
 - 4) *nonmemory task EEGs*: recorded when participants are performing a nonmemory-related task (e.g., a visual tracking task).
- 5) Second, we plot the performances separately for the following three different validation approaches.
- 6) *Validation with independent dataset (No overlap)*: the validation is implemented on an additional group of participants whose data are not used in training or tuning classifier parameters [see Fig. 1(a)].
 - 7) *Validation without independent dataset (No overlap)*: the validation is implemented on the same group of participants. However, the data for the training or tuning of classifier parameters and testing are independent (e.g., half of the data are used for the training of classifier parameters, and the other half of the data from the same group are used for testing) [see Fig. 1(b)].
 - 8) *Validation without independent dataset (overlap)*: the validation is implemented on the same group of participants, and data for the training or tuning of classifier parameters and testing are NOT fully independent (e.g., data used for training could later be included in the testing) [see Fig. 1(c)]. Fig. 1(d) further plots the classification accuracies against sample size (mean and standard deviation) for each combination of EEG states and validation approach.

Three key messages can be drawn from this figure. First, results based on a validation approach on nonindependent (overlap) data tend to show higher classification accuracies [see Fig. 1(c) and (d)]. However, due to the problem of data leakage between classifier training and testing, these results are prone to suffer a false inflation of classification performance. Second, with the most rigorous validation approach, resting-state EEGs

recorded after the task seem to provide better classification performances [see Fig. 1(d)]. However, only two papers with a relatively small sample size fall into this category. More studies will be needed to verify its advantages. Third, the average sample sizes for all combinations are less than $N = 150$, which shall be further improved if we want to increase the robustness of the results. Therefore, to improve the generalizability of findings, we apparently need more studies based on a validation method with a much larger sample size of nonoverlap independent data samples.

Based on a study that used both eyes-closed resting-state EEG and ERP, EEG provided critical frequency-based biomarkers for early-stage AD detection, particularly in the theta, beta, and alpha bands [91]. However, task-based EEG features derived from ERP, particularly the P300 component from auditory stimulation in the Pz channel, have been shown to yield better classification accuracy than EEG features alone [5]. Nevertheless, the referenced study did not utilize state-of-the-art algorithms in the field, such as the analysis of periodic components of EEG [92], which may outperform P300 components of ERP in distinguishing early MCI. Further research is warranted to directly compare these approaches and establish their relative utility in clinical settings.

In summary, both resting-state and task-state EEGs have the potential to serve as valuable neural markers for classifying between individuals with MCI and HC. Resting-state EEGs offer advantages in terms of ease of data collection and the assessment of global brain conditions, while task-state EEGs can provide more targeted information about specific cognitive impairments. The choice of approach should be guided by the specific research question, the target population, and the available resources, with the ultimate goal of developing sensitive and reliable neural markers to support early detection, diagnosis, and monitoring of cognitive decline. Furthermore, future research may focus on integrating these two approaches, exploring the complementary information they can provide, and developing comprehensive EEG-based assessment protocols that can enhance the early detection and monitoring of cognitive impairment.

V. SIGNAL PROCESSING AND EEG NEURAL MARKERS

Two key interconnected EEG or ERP patterns have been observed in MCI or AD compared to HC: 1) slowing of EEG activity; and 2) decreased connectivity or synchrony between brain regions [8], [74]. Slowing of EEG refers to an increase in lower frequency bands (e.g., theta) and a reduction in higher frequency bands (e.g., beta), often detected using power spectrum-based or nonlinear analysis approaches [8], [74]. Decreased connectivity in EEGs for MCI, particularly between the frontal and occipital lobes in the alpha band, reflects disrupted neural communication and impaired cortical network integrity [8], [93]. These patterns provide critical insights into the neurophysiological changes underlying cognitive decline. This section will discuss different methods used to detect such EEG and ERP changes in cognitive impairment. By analyzing these features, we can reveal significant information about the brain's functional changes

associated with MCI, thus enhancing our understanding of the complex neural mechanisms and patterns involved in cognitive impairment.

Building on the observed EEG patterns, their relationship with cognitive performance reveals significant disruptions in cortical and thalamocortical networks that are critical for memory and executive function [8], [74], [93], [94]. Elevated theta power is associated with impaired inhibitory control and reduced neural efficiency, correlating with poorer cognitive performance. Similarly, reduced alpha power in posterior regions indicates weakened cortical synchronization and disrupted information gating, which is critical for maintaining attentional and memory networks. The slowing of alpha peak frequency serves as a robust marker of neuronal dysfunction, signaling reduced oscillatory integrity needed for effective memory encoding and retrieval [8], [74], [93], [94]. Furthermore, the decline in beta power, crucial for higher-order cognitive processes such as working memory and sensory integration, indicates impaired synaptic connectivity and neural plasticity. These neurophysiological disruptions in MCI patients are strongly linked to pathological changes, such as amyloid-beta plaques and tau tangles, which interfere with synaptic transmission and network synchrony [8], [74], [93], [94].

A. Power Spectrum-Based Features

Examining the distribution of brain electrical activity across different frequency bands offers valuable insights into the brain's functional integrity [8], [92]. Here, we explore how power spectral analysis methods shed light on the complex landscape of EEG in MCI. Key methods of analyzing EEG in the spectral domain include the following.

Power spectral density (PSD): PSD quantifies the power within each frequency band, providing a detailed power distribution across the spectrum [8], [92], [95], [96], [97], [98], [99]. By analyzing the PSD function, researchers can go beyond simple ratios and uncover subtle variations in power within each band. This fine-grained analysis can potentially reveal crucial information that might be missed by more traditional methods. Research suggests that MCI patients might show a decrease in the power of higher frequency bands (beta and gamma), which are linked to complex information processing and cognitive flexibility [100], [101].

Relative PSD: While PSD offers detailed power distribution, the relative power spectrum gives information on the proportional contribution of each frequency band to the total EEG power. This method provides a high-level perspective on brain activity, showing potential disruptions in MCI. Researchers observed that a decrease in higher frequency bands and an increase in slower frequencies were associated with cognitive decline [100], [102], [103], [104], [105].

Peak/median frequency: These measures offer a simplified overview of the dominant frequency within an EEG epoch. A decrease in peak frequency or a shift toward a lower median frequency in MCI can indicate a decline in the brain's ability to generate faster, more intricate electrical activity patterns [102], [106].

Wavelet feature analysis: The EEG constantly presents non-stationary spectral characteristics [9]. Wavelet analysis provides

information on the nonstationarity of EEG signals by decomposing them into time–frequency components. This technique is valuable for identifying transient changes in brain activity associated with specific cognitive tasks in MCI patients [103], [107], [108].

Periodic and aperiodic components: EEG power spectra generally comprise two primary elements: an aperiodic background (often referred to as the arrhythmic or $1/f$ -like component) and periodic or rhythmic neural oscillations [109], [110], [111]. The aperiodic component, sometimes called fractal or “scale-free” activity, is characterized by its self-similarity across various temporal scales [109]. Research has highlighted the advantages of focusing on the periodic components of EEG, which are linked to cognitive functions such as processing speed and working memory [92], [110]. By excluding the aperiodic components, one can avoid potential masking effects that might obscure reductions in genuine oscillatory power, shifts in oscillation frequency, or decreases in broadband power, which are especially subtle in mild conditions such as MCI [92]. Specifically, it was shown that the EEG periodic components, based on the beta/theta power ratio over the occipital lobe, are a promising neurophysiological metric for distinguishing patients with MCI and AD from healthy individuals, warranting further study [92].

B. Connectivity

EEG/ERP connectivity analysis examines the functional interactions between different brain regions. It provides insights into the network dynamics and communication patterns essential for cognitive processing [112]. In the context of MCI, alterations in EEG connectivity can reveal brain circuit disruptions that underlie cognitive deficits [113]. This section discusses current approaches to assessing EEG connectivity, which can be broadly classified into two types: functional and effective connectivity.

Functional connectivity: Functional connectivity refers to the statistical dependencies between EEG signals recorded from different brain regions. It does not imply direct causality but rather indicates synchronous activity. Functional connectivity is commonly assessed using several methods:

Phase synchronization index: This coefficient quantifies the linear relationship between the amplitude envelopes of EEG signals from different regions. Reduced Pearson correlation in MCI indicates weakened synchronization and connectivity [114], [115].

Pearson correlation coefficient: This coefficient quantifies the linear relationship between the amplitude envelopes of EEG signals from different regions.

Amplitude envelope correlation: This method assesses the correlation between the amplitude envelopes of EEG signals, reflecting the comodulation of signal power. Altered amplitude envelope correlation in MCI patients can indicate disrupted communication within neural networks [114].

Graph analysis: Graph analysis leads to construct network graphs based on connectivity measures, from which it is possible to study analyzing properties such as node degree, clustering coefficient, and path length. Graph metrics can reveal global and local network disruptions in MCI, providing a comprehensive view of connectivity alterations [114]. For example, various studies showed that the small-world index, which measures the

efficiency and integration of brain networks by balancing local clustering and global connectivity, may be a useful biomarker for diagnosing AD and tracking the progression from MCI to AD [116].

In 2019, Franciotti et al. [117] compared the number of edges, inward edges, and outgoing edges through graph theory analysis of EEG from HC, MCI, and AD patients with mild dementia. Their results confirmed the hypothesis of an affected pattern of information flow in the brain networks. More recently, in 2021, Abazid et al. [118] analyzed a dataset of EEGs of subjective cognitive impairment patients, MCI, and AD patients. They quantified the graph links by weighting them by an entropy measure. They demonstrated the higher effectiveness of the entropy measure in analyzing the brain network in patients with different stages of cognitive dysfunction. Finally, Zhang et al. [119] classified MCI patients and normal controls using a two-layer random forest (RF) approach. They showed that the network metrics can provide useful information to assist in diagnosing MCI patients. Moreover, these authors observed a significant correlation between global network metrics and clinical cognitive evaluations. They concluded that graph theory analysis could act as a strategy to differentiate MCI patients from normal control subjects.

Effective connectivity: Effective connectivity describes the causal influence that one brain region exerts over another. It indicates directed interactions within the brain network. Effective connectivity provides insights into directed interactions within the brain:

Granger causality: Granger causality assesses whether one time series can predict another, implying a directional influence. Reduced Granger causality in MCI patients indicates impaired directed connectivity, particularly affecting information flow between critical brain regions [105], [115].

Direct transfer function (DTF): The DTF is a sophisticated method used in EEG signal processing to analyze the directional influence and connectivity between different brain regions. It quantifies the directional flow of information, with MCI patients often exhibiting reduced DTF values, indicating weakened directional influences between critical brain regions, particularly within the default mode network (DMN) [105], [115], [120].

Notably, directed connectivity is typically important for task-state EEG or ERP as it reveals the causal relationships between brain regions involved in cognitive processes. For instance, traveling waves of theta- and alpha-band oscillations propagate in a posterior-to-anterior direction during successful memory encoding and in an anterior-to-posterior direction during memory recall, dynamically coordinating neural connectivity across the cortex [121].

C. Complexity and Irregularity

In recent years, there has been a surge of interest in analyzing the complexity or irregularity of EEG and ERP signals for the diagnosis and monitoring of MCI, a condition often considered a prodromal stage of dementia. This section delves into the intricate world of EEG complexity and its potential role in deciphering the neurophysiological underpinnings of MCI. The

human brain is a symphony of interconnected neurons, generating EEG signals that exhibit a rich tapestry of rhythms and patterns. This inherent complexity reflects the brain's dynamic interplay between information processing, functional integration, and cognitive flexibility [122]. Unveiling this complexity is available through the utilization of the following complexity measurements.

Shannon entropy: Shannon entropy measures the uncertainty of an EEG signal. Higher entropy signifies greater irregularity or complexity, suggesting a richer repertoire of information content within the signal. In the context of MCI, studies have reported a decrease in Shannon entropy, potentially reflecting a decline in the brain's ability to generate diverse and informative neural activity patterns [123].

Spectral entropy: EEG signals comprise a spectrum of frequencies, each associated with specific cognitive functions. Spectral entropy quantifies the distribution of these frequencies within the signal. A broader frequency distribution, reflected by higher spectral entropy, indicates greater complexity. In MCI, a reduction in spectral entropy has been observed, suggesting a potential narrowing of the utilized frequency bands, which may be linked to impaired cognitive processing [102].

Permutation entropy and sample entropy: These techniques delve deeper into the sequential structure of the EEG signal, focusing on the order of data points. Permutation entropy measures the complexity of these ordering patterns, with higher values indicating more intricate and unpredictable sequences. Sample entropy, another entropy measure variant less susceptible to noise, provides similar insights. Studies suggest that both permutation and sample entropy decrease in MCI, potentially reflecting a decline in the brain's ability to generate dynamic and nonrepeating neural activity patterns [102], [124], [125].

Fractal dimension: The human brain exhibits remarkable self-similarity, where complex patterns repeat at different scales. The fractal dimension quantifies this self-similarity. A higher fractal dimension indicates a more intricate and interwoven structure within the EEG signal. Research suggests that the fractal dimension of EEG may be reduced in MCI, potentially reflecting a loss of this intricate neural organization [124].

Lempel-Ziv complexity: This method estimates the minimum number of symbols or patterns needed to describe the EEG signal effectively. A higher Lempel-Ziv complexity signifies a more intricate and information-rich signal. Studies have shown a decrease in Lempel-Ziv complexity in MCI, potentially reflecting a decline in the brain's ability to generate diverse and unpredictable neural patterns [102], [105], [124].

D. Others

Microstates are quasi-stable periods of brain activity represented by distinct, instantaneous scalp potential topographies [130]. Four EEG microstate classes, characterized by distinct functional associations and topographical orientations, have been identified: Class A (left posterior to right anterior orientation, correlated with verbal/phonological processing), Class B (right posterior to left anterior orientation, linked to visual processing), Class C (anterior-posterior orientation, associated

with autonomous processing), and Class D (central maximum with midline frontal topography, connected to attention-related processing) [130]. These classifications help in understanding the functional coordination within large-scale cortical networks and their relevance to various cognitive and neuropsychiatric conditions. Individuals with MCI also exhibited significantly higher occurrence and coverage of microstate A in comparison with HC, although microstate D may indicate underlying pathological changes related to the progression from MCI to AD [131].

Researchers also explored statistical features extracted from EEG signals, such as mean, standard deviation, and variance. Increased variance in specific frequency bands might indicate a loss of consistency in neural activity patterns, reflecting cognitive decline [107], [114], [115].

Advanced techniques such as independent component analysis (ICA) and principal component analysis decompose the EEG signal into its constituent components, revealing hidden patterns or activity sources that may be masked by the overall signal [98], [107], [123], [124], [125], [126], [128]. ICA, for example, can identify components associated with specific cognitive processes altered in MCI patients.

Waveform analysis methods, tailored to focus on specific aspects of the EEG signal relevant to sleep and MCI, offer insights into brain activity during sleep, a state often linked to cognitive decline. By combining sleep waveform features with spectral and complexity features from wakefulness, researchers can create a comprehensive picture of brain function in MCI patients [124].

One approach to promote the compensatory mechanisms in PFC is by enhancing executive function and working memory [39], [132]. In this context, analyzing EEG data from PFC while doing these tasks offers valuable insights. For example, the literature suggests the role of cross-frequency phase-amplitude coupling [theta phase–gamma amplitude coupling (TGC)] in working memory both in animals and humans [133], [134], [135]. Theta–gamma coupling is the synchronization of theta (4–8 Hz) and gamma (30–50 Hz) brain oscillations and is commonly observed in the hippocampus, PFC, and posterior brain areas during memory and learning tasks. Theta oscillations are linked to memory formation and spatial navigation [136], [137], while gamma oscillations are associated with information processing, memory, and attention [138], [139]. This coupling is thought to coordinate neural activity across brain regions during cognitive tasks, with theta oscillations timing the firing of gamma oscillations to aid in memory encoding and retrieval. Indeed, TGC reflects the coordination of neural networks involved in memory processes, contributing to functions such as ordering and manipulating information [133], [140]. It has been shown that TGC measured from PFC is associated with working memory in healthy younger adults [135], [141], healthy older adults [135], [141], [142], [143], and MCI patients [129]. This association was robust even after controlling for structural changes in MCI patients [144] and throughout a 12-week follow-up period in healthy older adults [142]. TGC could serve as a neurophysiological marker better at identifying individuals with MCI who are at high risk for cognitive decline or dementia

TABLE I
EEG/ERP FEATURES FOR THE DETECTION OF MCI

Category	Feature type	References
Frequency features	Relative power spectrum	[100],[102],[103] [104], [105]
	Power spectrum density	[97],[98],[100],[101],[103],[106],[107], [114],[120],[124],[126],[127]
	Wavelet feature	[103],[107],[108]
	Peak/median frequency	[102],[106]
Complexity	Permutation entropy	[124],[125]
	Sample entropy	[102],[124]
	Spectral entropy	[102]
	Shannon entropy	[123]
	Fractal dimension	[124]
	Lempel-Ziv complexity	[102],[105],[124]
Connectivity	Phase synchronization	[114],[115]
	Pearson correlation	[114],[115]
	Amplitude envelope correlation	[114]
	Graph analysis	[114],[124]
	Directed transfer function	[105],[115],[120]
	Granger causality	[105],[115]
Other	Microstates	[132]
	Statistical features	[107],[114],[115]
	Signal Decomposition	[98],[107],[123],[124],[125], [126],[128]
	Waveform	[124]
	Theta phase – gamma amplitude coupling	[129]

compared to clinical diagnosis alone [145]. Furthermore, TGC is lower in MCI patients compared to healthy older adults, despite these MCI patients showing no statistically significant impairment in performing the working memory task. These findings suggest the role of TGC in cognitive compensation, which would have to drop below a certain level before behavioral impairment becomes noticeable [129].

Table I summarizes the MCI-related EEG features. By integrating these diverse EEG analysis methods, researchers aim to develop improved diagnostic tools, enhance monitoring strategies, and better understand the neurodegenerative processes underlying MCI. In addition, we include a comparative table (see Table II) that contrasts different EEG/ERP analysis methods based on their relative strengths, weaknesses, computational load, and clinical relevance in MCI detection.

TABLE II
COMPARATIVE OVERVIEW OF KEY EEG/ERP FEATURE EXTRACTION METHODS FOR MCI DETECTION

Method	Concept	Strengths	Weaknesses	Computational Load	Clinical Relevance
Power Spectral Density	Identifies changes in brain activity across frequency bands	- Widely used and easy to interpret - Robust for band power analysis	- Misses time domain features - Can be sensitive to artifacts	Low	Beta–theta ratio shown promise in MCI detection [89], [139]
Wavelet Analysis	Decomposes EEG into time–frequency to capture nonstationary/transient events	- Good for transient event detection - Suitable for ERP-based studies	- Requires wavelet selection - Larger feature space	Moderate	Useful for differentiating subtle MCI patterns [104],[105]
Connectivity	Evaluates relationships among EEG channels (e.g., phase sync, Granger causality, DTF)	- Reveals network-level changes - Can highlight causal interactions	- Interpretation is complex - Noise and data-size sensitive	Moderate	Captures disrupted networks in MCI [102],[112],[113],[114],[115]
Complexity Analysis	Uses entropy/fractal measures (e.g., sample/permutation entropy) to quantify signal irregularity	- Sensitive to subtle neural changes - Complements linear/spectral approaches	- Parameter selection critical - Can be artifact-prone	Varies	Identifies early MCI changes via irregularity metrics [99],[117],[118]

E. Comparative Analysis of EEG Potential Biomarkers for MCI Detection

Meghdadi et al. [113] investigated various EEG-based methods to identify biomarkers of cognitive decline in MCI, focusing on both power spectrum and connectivity measures. They compared traditional power spectrum metrics such as the theta-to-alpha ratio (TAR) and coherence with their novel power distribution distance measure (PDDM), which captures temporal power distribution variations rather than averaging them. Their findings revealed that while TAR effectively detects significant changes in later stages of cognitive decline, PDDM demonstrates enhanced sensitivity for early detection of MCI-related changes, particularly in temporal regions, making it a promising tool for early-stage diagnosis. In addition, Smailovic et al. [146] found that increased theta and delta global field power were significantly correlated with decreased CSF amyloid beta 42, while decreased alpha and beta GFP were linked to increased p- and t-tau, demonstrating the sensitivity of EEG GFP measures in differentiating MCI from HC. In another study, Azami et al. [92] used various periodic and full-spectrum EEG features to distinguish HC from MCI, identifying periodic-based beta/theta ratios in the occipital region as the most effective biomarker for this differentiation [92].

Nevertheless, there remains a critical need for a comprehensive study comparing all state-of-the-art algorithms in the field of EEG and ERP, which would be invaluable for advancing future research and applications.

F. Interventions in Cognitive Impairment

Insights into functional connectivity and power spectral characteristics also provide a foundation for targeted interventions in cognitive impairment. Disruptions in thalamocortical circuits and the DMN, commonly observed in conditions such as MCI and AD, can be addressed through neurostimulation techniques [8]. Transcranial direct current stimulation (tDCS), for example, has been shown to modulate connectivity and power spectra by inducing cortical excitability changes. Anodal tDCS of the posterior parietal cortex has demonstrated increased alpha and

beta power, alongside transient theta band enhancements during stimulation, reflecting improved neuronal dynamics [147]. Similarly, high-frequency repetitive transcranial magnetic stimulation (rTMS) applied to the dorsolateral PFC has been shown to modulate electroencephalographic functional connectivity while improving cognitive function in AD patients [148]. Increased beta connectivity may play an important mechanistic role in the therapeutic effects of rTMS [148].

G. Power Spectrum, Entropy, and Connectivity Analysis in HC Versus AD

This section presents visual illustrations of key EEG signal characteristics in cognitive impairment, including power spectrum, entropy, and connectivity (analyzed using Granger causality). Fig. 2 presents a comparative analysis of EEG signals between an HC and an individual with early-stage AD (the EEG is very similar to one with MCI). The data were obtained from a publicly available dataset [149]. The HC is a 73-year-old female with a mini-mental state examination (MMSE) score of 30, while the AD subject is a 73-year-old female with an MMSE score of 23. The first subplot displays the time-domain EEG signals with length 2 s (sampling frequency = 250 Hz). The second subplot illustrates the PSD in the 1–30 Hz frequency range, showing notable variations in spectral power between HC and AD, particularly in the alpha and beta bands. The third subplot provides a bar graph comparing spectral power across the delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz) bands, indicating a reduction in alpha power for the AD subject. The nonlinear dispersion entropy (DispEn) algorithm [150] was also used to quantifying differences in signal irregularity, with the HC (DispEn = 2.61) showing higher DispEn compared to the AD (DispEn = 2.48). Overall, this figure and entropy values effectively highlight distinctions in spectral and entropy-based features between HC and AD EEG signals.

Fig. 3 illustrates the Granger causality analysis of directional connectivity between the frontal (F) and occipital (O) regions in EEG signals, using the same dataset as in Fig. 2. Using a lag of 12 samples (50 ms), the causality from F to O is significantly stronger in the HC (15.20) compared to the AD subject (11.95).

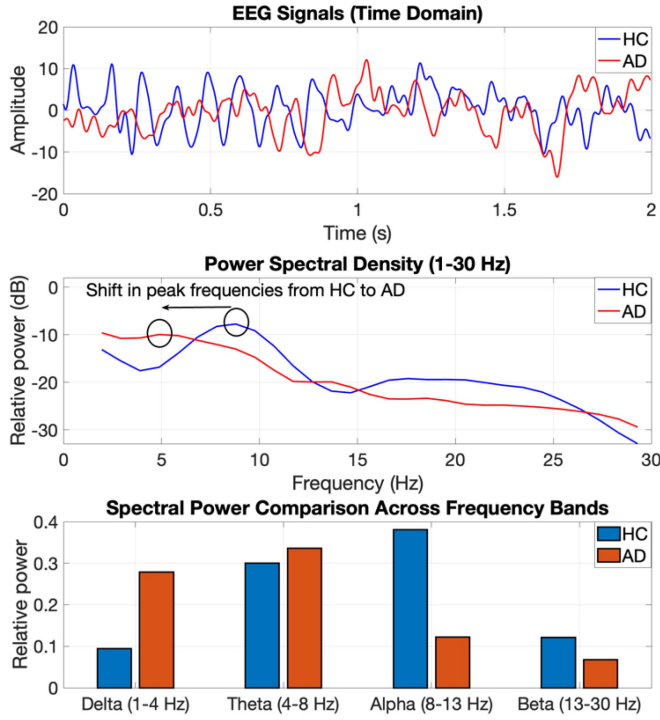


Fig. 2. Comparative analysis of EEG signals between an HC and an individual with early-stage AD. (Top subplot) Time-domain EEG signals. (Middle subplot) PSD comparison between HC and AD. (Bottom subplot) Bar graph comparing spectral power across the delta, theta, alpha, and beta frequency bands.

This result indicates diminished directional connectivity in AD, reflecting disruptions in neural communication pathways associated with the disease. The chosen lag aligns with the typical range of 1–12 samples (5–50 ms) used for capturing temporal dynamics in Granger causality models, considering the 2-s data length and the rapid changes inherent in EEG signals.

VI. MACHINE LEARNING-BASED CLASSIFICATION AND VALIDATION PROCEDURE

A. Classification Performance Index

Most of the collected studies used accuracy, sensitivity, and specificity to evaluate the classification performance of MCI versus control. These indexes can be directly drawn from the confusion matrix elements (TP: true positive, FP: false positive, TN: true negative, FN: false negative). Some studies only report *Accuracy* [151], [152]. Although *Accuracy* is easy to interpret, it ignores false positive and false negative rates, which play a more important role in clinical practice. This is why *Sensitivity* and *Specificity* are frequently used to replace *Accuracy* when assessing detection system performance. *Sensitivity* (or *Recall*) presents how well a classification model finds all instances of the positive class. Many studies also used the so-called area under curve (AUC) as a performance index [153], [154], [155], [156], [157], [158], [159], [160], [161], [162], [163], [164]. The AUC score can be obtained from the receiver operating characteristic (ROC) curve plot, which shows the trade-off between *Sensitivity* and *Specificity*.

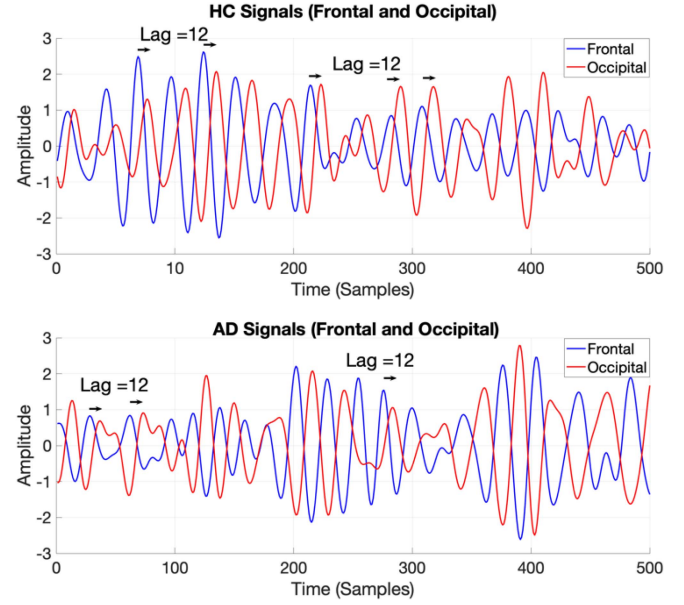


Fig. 3. Granger causality (lag = 12) analysis of EEG connectivity between the frontal (F) and occipital (O) regions for the same HC (top subplot) and AD (bottom subplot) individuals as in Fig. 2. Causality from F to O is stronger in HC (15.20) than in AD (11.95), indicating reduced connectivity in AD.

Patients with MCI are often more difficult to recruit than HC participants, which may explain why the datasets used in many previous studies are imbalanced: negative (control) data often outnumber positive (MCI). The F_1 score is an appropriate metric for assessing classification performance on an imbalanced dataset since it considers both *Recall* and *Precision*, where *Precision* answers the question of how many positive predictions are correct. A good MCI-HC classification model has a high F_1 score, meaning both *Recall* and *Precision* are high. The F_1 score was adopted in various studies [161], [165], [166], [167], [168], [169], [170], [171], [172], [173], [174], [175], [176], [177], [178]. There are also some other indices, such as Matthews correlation coefficient (MCC) and Cohen's Kappa value (k) [166], [172], [179], [180]. Although MCC might be more informative than accuracy and F_1 score [181], [182], it was adopted in only a few studies [183].

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Sensitivity/Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$F_1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

$$\text{MCC} = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (6)$$

$$k = \frac{2 * (TP * TN - FP * FN)}{(TP + FP)(TN + FP) + (TP + FN)(TN + FN)} \cdot (7)$$

B. Performance Evaluation Method

Three different approaches have been frequently adopted to predict the generalization performance of an MCI-HC classification in previous studies, including k -fold cross validation (k -fold CV), leave-one-subject-out cross validation (LOSO-CV), and train-valid-test split. One conventional approach to preparing data for the evaluation process is to consider each subject as an instance (subject-level). In this approach, the features of the segmented EEG epochs of one subject are calculated, and these features are further averaged across epochs. The averaged feature values from different electrodes are concatenated to one single vector, a data point. As a result, the number of data will be equal to the number of subjects. Another approach is to consider each EEG segment (epoch) of each subject as an instance (epoch-level), which, compared to the subject-level approach, results in a much larger sample size than the number of subjects.

In the k -fold strategy, all instances are divided into k parts, and in each iteration, one part is used as the test set, while the rest are used as the training set. In the EEG-based MCI detection literature, k -fold with $k = 5$ or 10 has been frequently used and can be applied at the subject-level [155], [156], [159], [161], [177], [184] or epoch-level [166], [172], [183], [184], [185], [186], [187], [188], [189]. It is worth noting that, in the case of epoch-level k -fold CV, each instance is not independent; different instances of the same subject are strongly correlated and may exist in both the training and test sets, which may lead to an overoptimistic assessment of classification performance due to personal information leakage [166], [172], [183], [187], [188].

Regarding LOSO-CV, in each iteration, the EEG data of one subject is used for testing, while the data of the remaining subjects are used as the training set, ensuring that the information from one subject is only utilized in either the training or the test set. LOSO-CV has been frequently used in subject-level approaches [152], [158], [168], [169], [190], [191], [192], [193], [194], [195], [196] and rarely at the epoch level [187]. A few studies used leave-one-epoch-out CV, where one epoch of one subject is used for testing, while the remaining epochs of that subject and all epochs of other subjects serve as the training set [189], [197], [198]. However, this is not a reasonable evaluation method because information leakage clearly occurs.

With the limited EEG dataset and the development of deep learning techniques, which require a large amount of data for training, the train-valid-test split strategy has become more popular. This method is mostly based on the epoch level, where most EEG instances are used for training and the remaining portion is used for validation and testing [151], [157], [160], [163], [165], [185], [199]. The validation set is utilized to optimize hyperparameters for machine/deep learning models, and then the performance of the model with optimal hyperparameters is validated on the test set. Several studies only split the dataset

into training and test sets; they can be at the epoch level [167], [173], [200], [201] or subject level [202].

C. Classification Model

Classifiers play a critical role in the cross-validation procedure. Many machine learning methods have been used to discriminate between MCI and HC groups. Examples include k -nearest neighbors (k -NN) [170], [171], [172], [177], [179], [183], [203], [204], [205], [206], [207], linear discriminant analysis (LDA) [159], [164], [172], [173], [192], [193], [194], [196], [197], [204], [208], [209], [210], quadratic discriminant analysis (QDA) [191], [194], [206], Bayesian classifier [159], [177], support vector machine (SVM) [156], [159], [161], [165], [166], [170], [171], [172], [174], [175], [176], [178], [183], [186], [191], [194], [195], [200], [206], [209], [211], [212], [213], [214], [215], [216], [217], logistic regression (LR) [154], [155], [156], [158], [159], [161], [162], [174], [190], [198], and RF [156], [169], [172], [175], [177]. It is unfair to compare those classifiers using the results in different studies because the datasets are different. Within the same datasets, some studies showed that SVM outperforms k -NN, LDA, and QDA [191], [194]. In [161], [198], SVM is superior to LR. SVM could be the best classifier in most studies. Its success should be attributed to its learning principle based on the structural risk minimization and the use of a kernel trick. Therefore, it can provide better generalization performance in MCI-HC classification.

VII. DEEP LEARNING

Traditional machine learning approaches have long been employed in detecting MCI using EEG or ERP data [8], [92], [93], [218], [219]. However, these methods often fail to achieve high classification accuracy when distinguishing MCI from HC [8], [92], [218]. To address this challenge, deep learning has emerged as a promising approach for detecting and extracting intricate patterns from complex EEG/ERP data, leading to superior performance [108], [200], [220], [221], [222], [223].

As a subset of machine learning, deep learning focuses on training multilayered neural networks to discern and extract nuanced or hidden patterns from large datasets [224]. This unique approach enables the early detection of MCI and AD by identifying subtle changes in brain activity that precedes clinical symptoms [200], [220], [221], [222], [223]. In addition, these models hold the potential to monitor the progression of cognitive impairment by tracking changes in EEG/ERP patterns over time. Furthermore, they enable personalized treatment by utilizing detailed patterns to tailor interventions and therapies to individual patients.

Traditional machine learning models, such as SVMs, RFs, and k -NN, rely on predefined feature extraction techniques, such as PSD, coherence, and entropy measures, to characterize EEG/ERP signals [8], [92], [93], [218], [219]. While these methods have been employed to detect MCI from EEG or ERP data, they often fall short of achieving high classification accuracy, particularly when distinguishing MCI from HC, because they rely on predefined features that may not fully capture the

hierarchical and spatiotemporal complexities of neural signals [8], [92], [200], [218], [220], [221], [222], [223].

As a subset of machine learning, deep learning models overcome these limitations by automatically learning feature hierarchies directly from raw or minimally processed EEG/ERP data [224]. In addition, by identifying subtle changes in brain activity that precede clinical symptoms, deep learning approaches enable the early detection of MCI and AD [200], [220], [221], [222], [223]. These characteristics allow them to capture complex spatiotemporal dependencies, such as phase synchrony across channels or dynamic frequency patterns, without requiring hand-crafted features. Furthermore, deep learning models can leverage transfer learning and pretrained architectures to improve performance on limited datasets, while techniques such as attention mechanisms enhance their interpretability by identifying task-relevant spatial and temporal patterns [97], [108]. These advantages enable monitoring cognitive impairment progression by tracking changes in EEG/ERP patterns over time. In addition, they support personalized treatment by leveraging detailed patterns to tailor interventions and therapies to individual patients.

Moreover, generative AI techniques, such as generative adversarial networks (GANs) and variational autoencoders (VAEs) [225], have a great and unique potential to generate high-quality semisynthetic EEG/ERP data [226], [227]. These models can create realistic samples that preserve the statistical properties of the original data while introducing variability, addressing the challenge of limited and imbalanced datasets often encountered in clinical studies. Semisynthetic data generation enables robust training of deep learning models, reducing overfitting and enhancing generalization to unseen data. For example, GANs can simulate diverse pathological EEG patterns associated with MCI, helping refine models to better distinguish between MCI and HC. In addition, generative AI supports data augmentation, domain adaptation, and anomaly detection, further enriching the diagnostic pipeline for early-stage cognitive impairment.

There is a significant performance gap between traditional machine learning methods and deep learning architectures in EEG-based HC versus MCI classification. For example, Kim et al. [108] demonstrated that the deep learning model CEEDNet outperformed traditional machine learning methods, including SVM and k-NN, in classifying EEG data into normal, MCI, and dementia groups. CEEDNet achieved a test accuracy of 74.66% on the CAUEEG-Dementia dataset, significantly surpassing the performance of SVM (52.33%) and k-NN (36.80%). The model's ability to detect subtle patterns and relationships within EEG data was highlighted by an impressive ROC-AUC score of 0.90 [9]. In another study, a multilayer perceptron framework classified EEG signals using features extracted from time-frequency and nonlinear phase-coupling (BiS) analyses. Compared to SVM and LR, the multilayer perceptron achieved superior performance, with an accuracy of up to 90.62% for HC versus MCI [228].

Similarly, it was shown that traditional classifiers such as SVM, LR, and k-NN achieved modest precision, recall, and F1-scores, with SVM slightly outperforming others for HC (precision: 0.74, recall: 0.86) but performing poorly for MCI (precision: 0.56, recall: 0.50). Conversely, deep architectures

such as the convolutional neural network (CNN) and convolutional autoencoder outperformed machine learning models, with CNN achieving an accuracy of 92% and effectively capturing complex temporal and spatial EEG features [229]. Another CNN-based model achieved 83% accuracy, 93% sensitivity, and 73% specificity in HC versus MCI classification, outperforming traditional machine learning models, where SVM recorded the highest F1-score of 0.53 for MCI [230]. These findings highlight the limitations of traditional machine learning methods in processing the nonlinear, nonstationary nature of EEG data and the superior capability of deep learning models to extract nuanced patterns, making them more effective tools for early MCI detection.

In the study of classifying MCI and HC subjects using EEG/ERP signals, various types of extracted features play a crucial role in enhancing the performance of deep learning models. These features often include complex transformations of the raw EEG signals to highlight significant patterns and differences [200], [220], [221], [222], [223], [231]. Commonly extracted features involve time-frequency representations, such as spectrograms, scalograms, and wavelet transforms. For instance, a continuous wavelet transform with different mother wavelets, such as the Mexican Hat or Morse wavelet, can generate detailed time-frequency graphs. Researchers then convert these graphs into 2-D images for input into a network such as a CNN [200], [220], [221], [222], [223], [231]. In addition, other studies have employed periodograms, representing the PSD of the EEG signals, and complex topographic maps that visualize the distribution of EEG activity across the scalp. These features capture temporal and spatial characteristics of the brain's electrical activity, facilitating the deep learning models to discern subtle differences among MCI and HC subjects. By using advanced architectures such as AlexNet [232], ResNet [233], and MobileNet [234], these studies have achieved high classification accuracies, indicating the effectiveness of these extracted features in differentiating between cognitive states [200], [220], [221], [222], [223], [231].

This review delves into various deep learning methodologies used for this task, including supervised, unsupervised (including self-supervised deep learning), semisupervised, transfer, and reinforcement learning. We will also discuss their respective advantages and disadvantages, providing insights into their effectiveness in MCI detection using EEG/ERP data.

A. Supervised Deep Learning

In deep supervised learning with neural networks, the network compares its output to a desired target value provided by a supervisor. This comparison generates an error signal, which can be calculated in various ways (for example, by squaring the difference between the output and the target value) [224]. The primary goal of training a deep supervised network is to adjust the internal connections (such as weights) to minimize the total error, which is the sum of all the individual errors. By achieving a low total error, the network ideally learns patterns that enable it to make accurate predictions on new, unseen data [224]. Three popular deep learning architectures used for

supervised EEG/ERP analysis are CNNs [200], [231], recurrent neural networks (RNNs), and long short-term memory (LSTM) networks [229], each with its own advantages and disadvantages. Recently, advanced deep neural network architectures, such as transformers have also enhanced the detectability of MCI from EEG data [235].

Convolutional neural networks: CNN is a deep learning architecture designed to process data with a grid-like topology, such as images, utilizing convolutional layers to automatically extract features by applying learnable filters, followed by pooling layers to reduce dimensionality [200], [229]. CNNs detect both local and global patterns, improving the model's ability to identify complex relationships in EEG/ERP signals, as well as spectral, complexity, and connectivity features extracted from EEG data [114]. However, CNNs require a substantial amount of labeled data to train effectively, and their training can be computationally expensive depending on the model architecture and training settings [200]. To address the challenge of limited labeled data, data augmentation techniques have proven effective in expanding the sample size of EEG data, thereby improving the detection of MCI [98].

Recurrent neural networks and long short-term memory: RNNs and LSTMs, on the other hand, take into account the long-range dependencies and temporal patterns of EEG/ERP, allowing them to analyze the temporal dynamics of brain activity, such as identifying patterns that unfold over time or segmenting the data into distinct stages or events [236], [237]. LSTMs, a special type of RNN, are specifically adept at capturing long-term dependencies within the data sequences, making them perfect for analyzing slow-wave activity or ERP components with extended time courses. Nevertheless, RNNs can suffer from vanishing or exploding gradient problems, making them difficult to train, especially for long sequences [224].

Hybrid model: The combination of CNNs and RNNs (or LSTMs) is also used to benefit from the advantages of both of them (hybrid models) [238]. In this way, CNNs are typically first used to extract spatial features from EEG/ERP data, which are then fed into RNNs to capture temporal dependencies [238]. This combination allows the model to deal with both the spatial and temporal information of EEG/ERP time series. However, such models are more complex than CNNs or RNNs and computationally expensive, requiring careful tuning and optimization. These models also require noticeable computational resources and large amounts of labeled data [224].

B. Unsupervised Deep Learning

Unsupervised deep learning, including self-supervised deep learning, proposes ways for training the weights in each layer in an unsupervised fashion, enabling the model to identify underlying structures and patterns within the data [224], [239]. While this approach can potentially reveal valuable insights from unannotated data, it often requires a substantial amount of EEG or ERP data and may yield less interpretable results [224], [240], [241], [242], [243]. As a form of unsupervised learning, self-supervised learning involves the system learning

to predict part of its input from other parts. Such approaches are especially beneficial in EEG/ERP applications with scarce labeled data. Autoencoders, including their extensions such as VAEs, GANs, and contrastive deep learning are the main types of unsupervised deep learning approaches trained to replicate their input as output [244].

Autoencoder: An autoencoder comprises an encoder that compresses the input into a lower-dimensional latent space and a decoder that reconstructs the input from this latent space. Such methods can be specifically used for feature extraction and anomaly detection in EEG/ERP data [229]. Autoencoder networks are also used for EEG data augmentation [238]. VAEs extend traditional autoencoders by learning a probabilistic latent space, enabling better generalization and generation of new EEG/ERP data samples [245], [246]. The primary drawbacks of these methods include the potential for overfitting the training data, particularly with complex models, and the challenge of interpreting the learned features. Masked autoencoder (MAE) is another type of self-supervised learning where some parts of the input data are masked, and the model is trained to reconstruct the missing parts [247]. Although it has not been applied to this application, the MAE-based model can learn meaningful patterns and dependencies within the EEG/ERP data. It is worth noting that such a model needs a careful design of the masking strategy to ensure effective learning.

Generative adversarial network : GANs, a type of unsupervised learning, consist of two neural networks: a generator and a discriminator. These networks are trained simultaneously in a competitive manner [248]. The generator generates artificial data, while the discriminator assesses its authenticity compared to real data. The objective is for the generator to produce data that are indistinguishable from real data. GANs present a promising tool for detecting MCI versus HC in EEG data by augmenting datasets with synthetic samples, facilitating unsupervised feature learning, enhancing discriminative performance through adversarial training, and capturing complex relationships inherent in EEG signals, thus potentially improving diagnostic accuracy and enabling early intervention in cognitive impairment disorders [226].

Contrastive Learning: Contrastive learning is a technique that involves training a model to differentiate between similar and dissimilar pairs of data points, which can be particularly useful in EEG/ERP analysis for learning robust representations of brain activity [241], [249]. These models are typically trained using contrastive loss functions, where the goal is to maximize the similarity between positive pairs (similar data) while minimizing the similarity between negative pairs (dissimilar data) [241], [249]. By doing so, these approaches can effectively learn discriminative features that distinguish between different cognitive states, such as HC and MCI groups. However, it is important to note that the success of contrastive learning hinges on carefully selecting positive and negative pairs. This task can be challenging due to the complexity and variability of EEG/ERP data. Nevertheless, with proper attention to pair selection, contrastive learning can significantly enhance our understanding of cognitive states and improve diagnostic accuracy in cognitive impairment disorders.

C. Semisupervised Deep Learning

Semisupervised deep learning approaches combine a small amount of labeled data with many unlabeled data during training [250], [251]. These methods offer some advantages in MCI/AD detection, including improved utilization of unlabeled data, enhanced model generalization, and reduced annotation costs. However, these methods also come with certain limitations, such as the complexity of model training, the risk of potential overfitting, and the requirement for extensive computational resources [250], [251].

D. Transfer Learning

Transfer learning in deep learning, which can be regarded as both a form of supervised and unsupervised learning, involves utilizing a pretrained model on a related task and fine-tuning it for a new but related task [252]. In supervised transfer learning, a model initially trained on a labeled source task transfers its knowledge to a related target task with limited labeled data, while unsupervised transfer learning involves transferring knowledge from an unlabeled source domain to a target domain, aiding learning when labeled data is scarce or absent [252]. For instance, pretrained deep learning models on a large, unrelated dataset can be fine-tuned for the specific task of EEG classification (AD versus MCI versus HC), requiring less EEG or ERP data compared to training a CNN, which is crucial given the expense and time-consuming nature of collecting large, high-quality EEG/ERP data [253]. This approach also reduces computational resources and learning time, enhancing model performance by adapting existing knowledge to new contexts. However, fine-tuning a pretrained model for EEG data necessitates careful selection of layers and hyperparameters to mitigate overfitting and ensure robust performance on unseen data.

E. Reinforcement Deep Learning

Reinforcement learning, which involves training an agent to make sequential decisions by rewarding good actions and penalizing bad ones, has not been explored in MCI detection using EEG or ERP signals, to the best of our knowledge [254], [255]. However, it holds promise for this application. Several reinforcement learning algorithms, such as Q-learning and deep Q-networks, could be applied to MCI detection using EEG data. These methods offer adaptive learning and real-time analysis capabilities, potentially enhancing feature selection for diagnosis [254], [255]. Integration with deep learning architectures, such as CNN and RNN, could further improve decision-making abilities in MCI studies. However, challenges such as computational complexity and data requirements may limit their utility in clinical settings, specifically for MCI diagnosis using EEG/ERP signals [254], [255].

VIII. FUTURE WORK

As we advance in our understanding and analysis of EEG and ERP data in MCI, several promising avenues for future research emerge. Addressing these areas will enhance our ability to diagnose and understand brain activity in MCI. The following key directions are particularly noteworthy.

A. Multivariate Multiscale Entropy

Multivariate multiscale entropy methods offer a robust framework for analyzing the complexity and dynamism of EEG and ERP signals [256], [257]. In the context of MCI, applying such methods can reveal how neural complexity changes across different scales of both the time and spatial domains. By examining the multivariate multiscale entropy of EEG or ERP signals recorded from multiple electrodes, researchers can assess the degree of signal complexity and its variations across different frequency bands. Integrating multivariate multiscale entropy methods with advanced spatial techniques, such as source localization or connectivity analysis, can provide a comprehensive view of the brain's functional architecture and its changes in MCI, leading to potential new biomarkers and insights into disease mechanisms.

B. Periodic Versus Aperiodic Components of EEG

As mentioned earlier, EEG power spectra consist of both periodic and aperiodic components, each reflecting different aspects of brain activity [109]. Periodic components, which are linked to rhythmic brain oscillations, are particularly relevant to cognitive impairment, while aperiodic components capture broader, less predictable aspects of neural dynamics [109], [154]. In MCI studies, distinguishing between these components can offer valuable insights into the neural changes associated with cognitive decline. Although the importance of periodic EEG components, compared to the well-known PSD, has been demonstrated in MCI [154], this finding needs to be replicated in larger datasets. Future research can also focus on leveraging the advantages of periodic components and their time-frequency representations as inputs for deep learning models.

C. Higher-Order Interactions

Despite its efficacy and popularity, functional connectivity approaches have limitations as they can only reveal pairwise dependencies, possibly overlooking more intricate, HOIs [258], [259], [260]. HOIs denote the complex relationships or dependencies among three or more units, elements, or variables in a system simultaneously that go beyond simple pairwise interactions [261]. Based on the definition of HOIs rooted from information theory, it is possible to distinguish between qualitatively distinct types of interactions: redundant interactions, where information is duplicated across multiple elements, leading to the resolution of uncertainty in one element when observing another; and synergistic interactions, where uncertainty can only be resolved by considering the joint state of two or more variables [258]. The exploration of this realm of redundant and synergistic interactions in the brain is limited because these interactions are typically beyond the reach of a bivariate functional connectivity network analysis [258]. To estimate the synergy- and redundancy-dominated interactions in a network of multiple elements or variables, O-information (OI) on the basis of mutual information (MI) has been recently introduced [262] and was also used for the application of AD [263]. The OI rate, the generalized form of MI rate (MIR, which shows how the MI between two variables changes over time), was

also introduced to assess Granger-causal effects, instantaneous influences, and extend these assessments into the frequency domain [264]. This framework utilizes the spectral representation of vector autoregressive and state space models to investigate how various groups of processes collaborate and duplicate their effects. This dynamic perspective is crucial for understanding cognitive processes and frequency-specific variations in brain activity, making MIR an essential analytical approach in EEG analysis [264]. In the context of MCI, exploring HOIs can provide deeper insights into how different brain regions interact and how these interactions change as the disease progresses. By examining HOIs, researchers can identify subtle disruptions in brain network dynamics that may be indicative of MCI, leading to the discovery of new diagnostic markers and a better understanding of the disease's impact on brain function.

D. Causal and Explainable Deep Learning

Deep learning approaches, while powerful, are often criticized for their lack of reproducibility and transparency, making them challenging to interpret and understand. To address these challenges, future research should focus on developing novel, explainable causal deep-learning techniques that can reveal nonlinear patterns and biomarkers in EEG and ERP data from individuals with MCI compared to HC. For example, researchers can explore the use of causal deep learning models integrating a structural dimension, which incorporates partial but testable causal knowledge, a parametric dimension that captures the type of relationships among variables, and a temporal dimension that models interactions over time [265]. These models help identify how changes in brain activity relate to cognitive decline in MCI.

To address the “black-box” nature of deep learning models, it is crucial to implement explainable AI techniques. SHapley Additive exPlanations can be used to attribute contributions of each feature to the model's predictions, offering insights into how different aspects of the data influence the outcomes [266]. Local interpretable model-agnostic explanations provide a way to approximate the model locally, making it easier to understand predictions for individual instances [267]. Attention mechanisms can also highlight which parts of the input data the model focuses on, offering further interpretability [268]. By integrating these explainable techniques, researchers can enhance the transparency and interpretability of deep learning models, ensuring that the insights derived from EEG and ERP data are actionable and clinically relevant. This approach will improve the reproducibility of findings and facilitate the identification of meaningful biomarkers and patterns in the progression of cognitive impairment.

E. Combining EEG With (f)MRI or fNIRS for Complementary Multimodal Analysis

Integrating EEG with MRI, fMRI, or fNIRS represents a powerful approach for obtaining a more comprehensive understanding of brain function in MCI. While EEG provides high temporal resolution and can capture rapid changes in neural activity, fMRI and (f)NIRS offer high spatial resolution and insights into hemodynamic responses associated with brain activity. Combining

these modalities can yield complementary information about brain dynamics. For example, it was shown that the combination of MRI and EEG provides a more comprehensive understanding of brain function and structure by integrating high-resolution anatomical details with real-time electrical activity, enhancing diagnostic accuracy and insight into AD [269].

Combining EEG and fMRI provides a comprehensive view of brain function by integrating the high temporal resolution of EEG with the high spatial resolution of fMRI, enabling a detailed analysis of how electrical brain activity correlates with hemodynamic responses across different brain regions [270]. Similarly, as fNIRS provides valuable information on cortical oxygenation and blood flow changes, a hybrid EEG-fNIRS model has shown superior classification accuracy for AD compared to using each modality alone, underscoring its potential to improve diagnostic and assessment processes for AD [271]. Future research should focus on developing methodologies for effectively integrating these data streams through multimodal fusion techniques or joint analysis frameworks, particularly emphasizing larger datasets and research on MCI rather than just AD.

By pursuing these research directions, we can advance our ability to diagnose, understand, and monitor MCI using EEG/ERP, ultimately leading to better patient outcomes and more effective interventions.

AUTHOR CONTRIBUTION

All authors conceptualized the direction and scope of this review article. Specifically, H. Azami contributed to Sections I, VII, and the organization of all sections. M. Mirjalili and T. K. Rajji contributed to Section II. Y.-H. Liu and H. Azami contributed to Section III. C.-T. Wu contributed to Section IV. C.-T. Wu and T.-T. Trinh contributed to Fig. 1. H. Azami contributed to Figs. 2 and 3. A. Humeau-Heurtier, T.-P. Jung, H. Azami, and C.-S. Wei contributed to Section V. Y.-H. Liu and T.-T. Trinh contributed to Section VI. T.-P. Jung, C.-S. Wei and H. Azami contributed to Section VIII. H. Azami, A. Humeau-Heurtier, C.-T. Wu, T.-P. Jung, and Y.-H. Liu further edit the final draft.

SUPPLEMENT

Individuals are diagnosed with MCI if they have the following criteria:

1. Evidence of modest cognitive decline from a previous level of performance in 1 or more cognitive domains reported by the patient, informant, or clinician (e.g., documented or observed decline over time).
2. The cognitive deficits do not interfere with capacity for independent day-to-day activities.
3. The cognitive deficits are not present exclusively in the context of a delirium.
4. The cognitive deficits cannot be better explained by another mental disorder.

Tests such as the Montreal Cognitive Assessment (MoCA) [272] and Mini-Mental State Examination (MMSE) [273] are widely used to evaluate various cognitive functions, including memory, attention, language, and visuospatial abilities, and to

diagnose MCI. Specific designated tasks can assess each cognitive domain separately. Clinicians typically employ a composite z-score to score performance on each task within the domain. For example, tasks like the N-back [274] and Sternberg [275] paradigms can be utilized to assess working memory. In the N-back task, participants are presented with a sequence of stimuli and must indicate whether the current stimulus matches the one presented N steps earlier. This task challenges individuals to maintain and update information over time. Similarly, the Sternberg task involves presenting participants with a set of items to remember, followed by a probe item, and participants must determine if the probe item was part of the original set. These tasks evaluate working memory by measuring the ability to temporarily hold information and comparing items held in memory and new stimuli.

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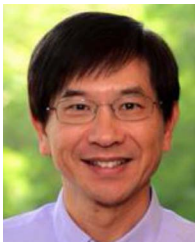
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